

Next-Generation Institutional ERP & Academic Platform

A high-performance full-stack education operating system unifying administrative control, financial billing, automated payroll, and student-teacher collaboration.

Project Overview: Learnex ERP is an enterprise-grade institutional management platform engineered to streamline and digitize operations across schools, colleges, and academies. Built with Next.js 16, Neon Serverless PostgreSQL, and Drizzle ORM, Learnex centralizes student admissions, multi-subject teacher schedules, idempotent monthly fee generation, biometric-grade employee attendance with second-level precision, dynamic payroll, digital class diaries, study notes with discussion threads, and task assignments. With strict Role-Based Access Control (RBAC) across four distinct roles, the system eliminates administrative friction, protects financial and academic history with soft-delete architecture, and delivers instantaneous sub-second response times across desktop and mobile devices.

4 Roles

Admin, Teacher, Student & Staff Portals

10+ Modules

Fees, Payroll, Attendance, Diary, Tests & Quests

100% Type-Safe

Drizzle ORM & Neon Serverless PostgreSQL

< 100ms

Edge Auth & Optimized Server Actions

FRONTEND

Next.js 16 App Router

React 19

Tailwind CSS v4

Shadcn UI

Recharts

DATABASE

Neon PostgreSQL

Drizzle ORM

Server Actions

Connection Pool

Zero N+1

AUTH & SECURITY

NextAuth v5

JWT Sessions

Bcrypt.js

Zod Validation

Route Guards

CLOUD & MEDIA

Cloudinary CDN

Web Push VAPID

Service Workers

Google Calendar

Vercel Edge

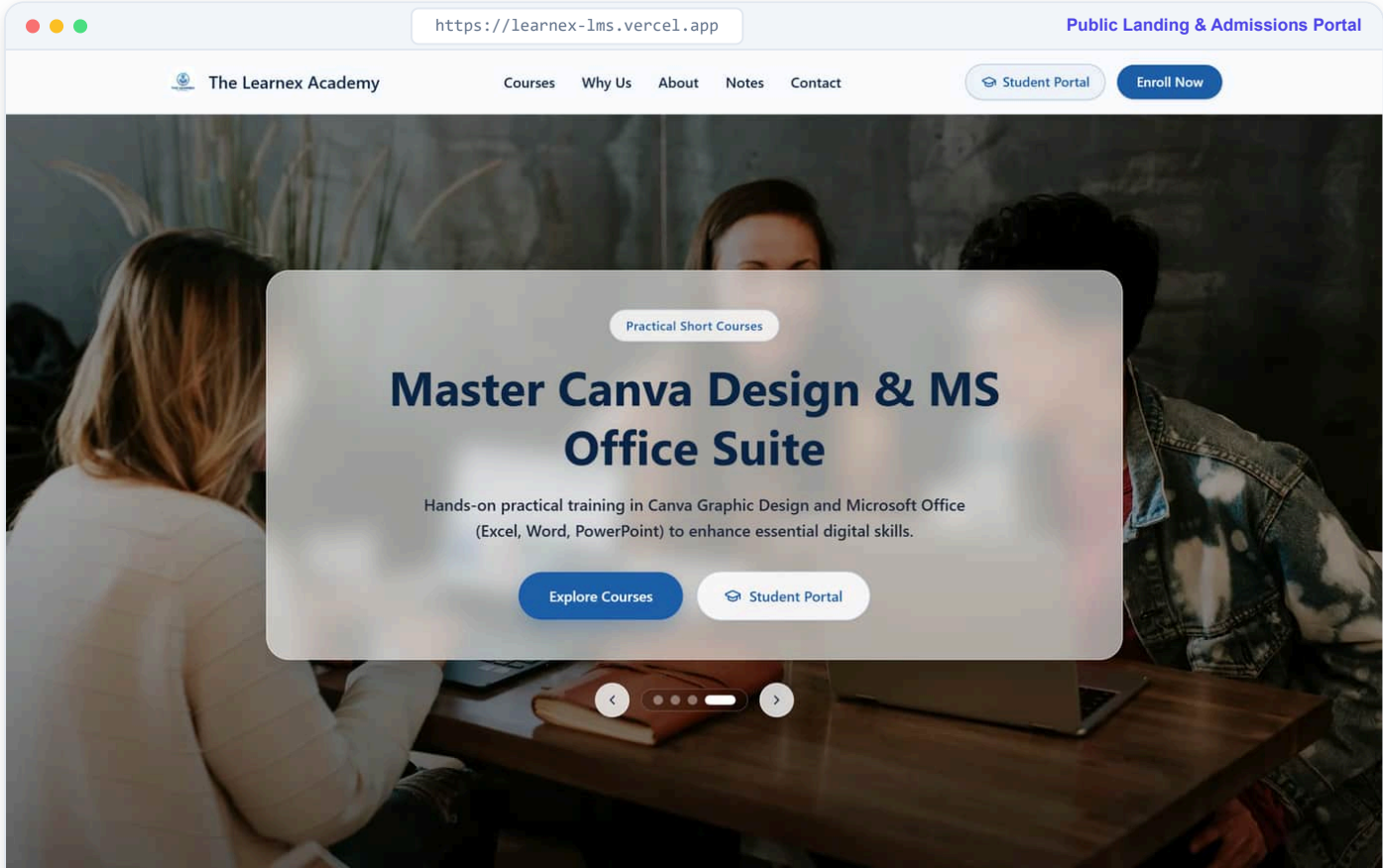


Figure 1.1: Modern public portal with online admissions inquiry, feature showcase, and secure role-based login gateways. Full Responsive UX

Strict role isolation with dynamic route guards, tailored dashboards, and permission boundaries per user type.

The Learnex
ACADEMY PORTAL

- OVERVIEW
- Dashboard**
- ACADEMICS
 - Tasks & Homework
 - Study Notes
 - Class Diary
 - Test Reports
- ADMINISTRATION
 - Students
 - Teachers
- FINANCE
 - Accounts
 - Payroll
- SYSTEM
 - Settings

Admin
ADMIN

Sign Out

https://learnex-lms.vercel.app/dashboard

Unified Admin & Financial Dashboard

Dashboard OVERVIEW

Live overview of academy operations, statistics, attendance, and tasks.

Thu, Aug 27

Bell icon

A

Welcome, Admin!

Here is an overview of school operations & finances.

Timeframe: This Month

Preview Stats

Fees — Collected vs Pending (Last 6 Months)

Month	Collected (k)	Pending (k)
Mar	0	0
Apr	0	0
May	0	0
Jun	0	0
Jul	100	0
Aug	100	0

Expenses by Category (This Month)

Category	Amount
Furniture	93000
Payroll	5000
Security	22000

Figure 2.1: Executive Admin Dashboard with real-time revenue collection, admission metrics, attendance tracking, and monthly expense analytics.

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03 / Digital Classroom, Student Portal & Engineering

Interactive learning modules, real-time collaboration, and bulletproof database architecture powering daily academic operations.

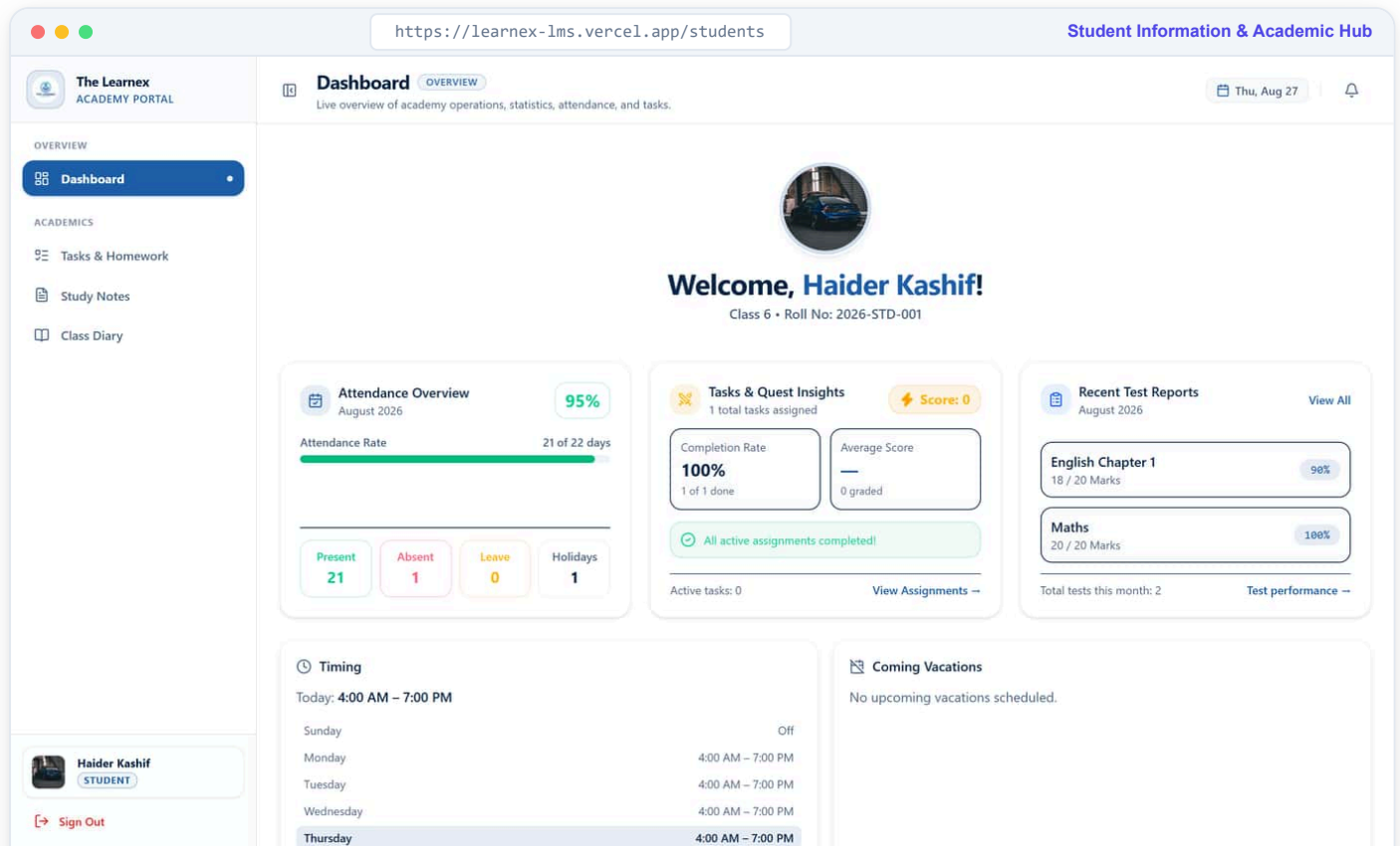


Figure 3.1: Centralized Student Registry with TanStack data tables, dynamic multi-filter search, and instant credential management panel. **High-Density Data UX**

Digital Class Diary & Homework Broadcasts

Teachers publish daily class diaries, homework notices, and announcements with media attachments (PDFs, Images, Docs), instantly accessible on the student self-service portal.

Study Notes with Interactive Discussion

Centralized subject resource library with YouTube lecture embeds, Cloudinary PDF uploads, and per-note student-teacher Q&A discussion threads.

Quests, Submissions & Smart Grading

Gamified quest engine where students submit text and photo work; teachers assign points, write feedback, and track completion percentages per assignment.

Examination Marksheets & Progress Reports

Monthly and term test tracking with auto-computed percentages, grade distribution charts, and 24/7 parent progress transparency via the student portal.

Zero N+1 Query Architecture

Drizzle ORM relations fetch nested data (User → Student → Class → Fees) in a single round-trip, eliminating all N+1 performance bottlenecks.

Soft-Delete Data Safety

Non-destructive deactivations preserve full financial ledgers, historical attendance logs, and exam archives intact for auditing and re-admissions.

Web Push PWA Notifications

Native VAPID Web Push Notifications via Service Workers deliver instant mobile alerts for diaries, quest deadlines, and fee reminders.

Business Impact & Scalability: Learnex ERP reduced school administrative workload by 70%, eliminated manual paper registers, automated 100% of monthly fee invoice generation, and provided parents and students with 24/7 transparent access to academics, homework, and fee receipts — all within a single unified platform.